Chapter 17

Market Access, Pharmaceutical Pricing, and Healthcare Costs

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Learning Objectives

Objective 17.1 Outline market access policies by pharmaceutical and medical technology industries.
Objective 17.2 Overview pricing practices of pharmaceuticals.
Objective 17.3 Explain types of healthcare costs and their estimation.

OBJECTIVE 17.1. OUTLINE MARKET ACCESS POLICIES BY PHARMACEUTICAL AND MEDICAL TECHNOLOGY INDUSTRIES

Market access (MA) can be defined as the set of strategies, activities, and processes that pharmaceutical (pharma) and medical technology (MedTech) companies develop to ensure that their products (drugs, devices, other technologies) are made available and adequately priced in a specific health system. MA supports the ability for a product to achieve at a fair reimbursed price and a favorable recommendation for prescribing through a health system.[1,2]

In this respect, MA is a multidisciplinary field that includes or requires contributions from various business functions, such as managing market channels, stakeholders, key opinion leaders (KOLs), pricing and reimbursement, government affairs, health economics and outcomes research, regulatory, and payer management. Therefore, it is a multidisciplinary field that includes or requires contributions from various other business functions, such as managing market channels, stakeholders, and KOLs.[1]

In the last 20 years, MA has gained considerable attention within the pharmaceutical industry as countries and health systems have started new policies to contain their escalating healthcare expenditures, especially during the last decade due to the global economic slowdown and crisis. Governments and health authorities have adopted stricter measures for new product approvals and introduced several guidelines and actions on clinicians to steer and limit their discretion in prescribing drugs and adopting technologies.

In this new scenario, pharma and MedTech companies have to cope with new challenges posed by decisions and influence exerted from various sources, including government agencies, top management of healthcare organizations, stakeholders, and KOLs.

The pharma and MedTech industries have traditionally built their success on the strategy of "share of voice." The more a company hits the clinicians with its field force, the more sales and revenues are expected. There is a correlation between coverage of the field force and sales; therefore, companies would adopt an aggressive push strategy to ensure their products succeed in the market.[3]

Introduction of a drug/device to the market includes at least three steps. First, a dossier with data on efficacy, safety, and tolerability is submitted to regulatory agencies to get an approval for the authorization of the introduction of the product in the market. Getting market authorization from the regulator is the responsibility of the medical/regulatory affairs division within a pharmaceutical company.

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Once the regulatory agency grants market authorization, the MA division prepares a dossier to be submitted for pricing and reimbursement. The national agency that approves price or reimbursement is most of the time different from the regulatory agency that provides market authorization. Sometimes the pricing and reimbursement agency could be independent of the government too but works in an advisory position to the government.

Third, the product is marketed to the targeted physicians, the prescribers, or those that can induce its consumption or utilization. In this stage, the marketing division and other parties within the company proceed with channels to make the product available to customers.

Thus, the introduction of a new product to the market, or revitalizing an old one, is a process involving a few stakeholders—governmental or nongovernmental agencies, pharmaceutical companies, physicians, and in some cases also pharmacies if authorized to prescribe or advise on the use of the product.

However, in the last two decades, the introduction of new products in the market has become a much more complicated and complex process. The MA landscape has evolved primarily due to escalating healthcare costs with difficulties to be reimbursed, safety concerns of newer drugs, aging population, and disease burden that exhaust limited drug/device budgets.

As the institutional environment gets more challenging, healthcare authorities are investigating more in-depth the claim for the value of new products; choices on pricing and reimbursement have become more cautious. For instance, reference pricing and generic substitutions are among the implemented (and accelerating) techniques used to anchor pricing and increase the affordability of new products. Furthermore, pricing or reimbursement agencies (governmental or not) are shifting from cost-sharing approaches to pay for performance (Fig. 17.1).

In this new scenario, pharma and MedTech companies have seen the emergence of a new and diverse set of stakeholders, and a brand new strategy for MA has become fundamental, i.e., specific tools and tactics are needed to support companies to strategize, plan, implement, and monitor engagement activities for the new stakeholders.

The intensity of this new landscape might vary quite a lot from mature markets, such as Europe and North America, consolidating and developing markets (e.g., Latin America, Russia, India, China, Middle East, and some African countries). However, in all these markets, new stakeholders are gaining dominant positions, generating the need for a new "share of voice."

Access and penetration of a market are no longer just a function of the relationship with clinicians and KOLs. Increasingly, MA requires good and deeper interactions with agencies/authorities, hospital managers, hospital pharmacists, leaders of governing authorities, HTA committee members, insurance managers, doctors involved in clinical governance, patient associations, and influential associations (such as "choosing wisely," "slow medicine," etc.).

Specifically, concerned agencies for MA are today a more sophisticated group of stakeholders that shape healthcare policy, establish the rules of conduct for pharmaceutical and MedTech companies to operate in the market, and sometimes are even the payers. Managers and pharmacists of hospitals are exerting greater control over clinician behaviors. Patients today are more aware of treatment modalities, effectiveness, and cost; patients can be very active and demanding, especially if they are charged with co-payments. Pharmacies are key stakeholders that could influence drug access by controlling the availability of the product in the retail or out-of-pocket market. Patient associations and other advocacy groups have the will and need to play an influential role in MA. The innovative new class of drugs needs to be marketed at a reasonable time and should be affordable (e.g., in rare diseases).

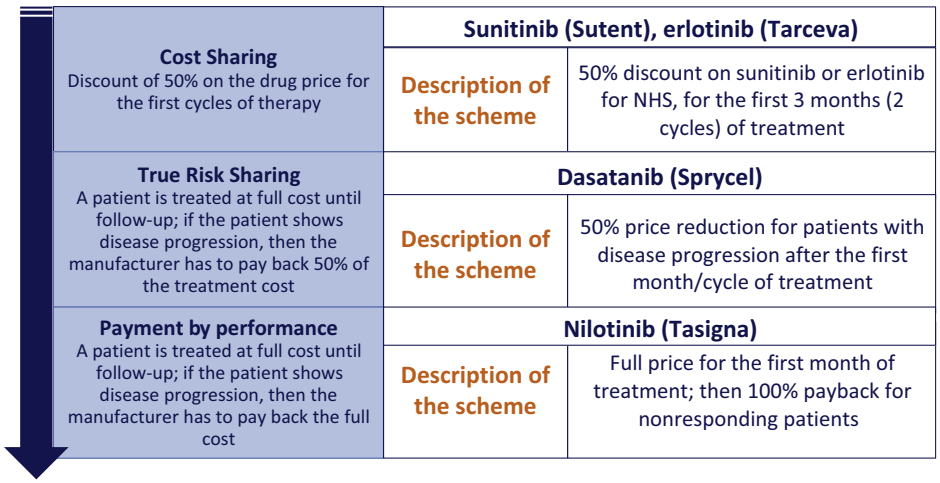


FIGURE 17.1 Evolving landscape in pricing: from cost sharing to pay for performance. *Prepared by using data from Italian Medicines Agency.*

In this scenario, physicians and KOLs are less dominant in the MA value chain. They still are a key "target"; their discretion in prescribing new and expensive products is limited because of

- growing austerity measures,
- the clinical governance tools operated by managers and pharmacists,
- the guidelines to be followed, and
- further restrictions set by health agencies.

The MA landscape in health systems has become quite complex, with intertwined and interlocked relationships among various stakeholders. Pharmaceutical and MedTech companies thus need to identify the right stakeholders that need to be engaged as part of the introduction of the product on the market and to identify the right and tailor-made actions for their engagement. This is why a new "share of voice" is needed.

The MA function is, therefore, the emerging part of the organization of most pharma and MedTech industries. A function that can also work as a link among the more traditional functions, such as R&D, sales, marketing, regulatory affairs, pricing and reimbursement, and medical affairs. MA would become the planner and coordinator of the efforts from all of them, helping the company to build on possible synergies and facilitate coordination in "talking" to stakeholders, rather than having different functions aimed at "share of voice" in an isolated way, with each one talking with multiple stakeholders with probable confusing and mixed messages.

When introducing a new product on the market, a company should do the following:

- consider the implications on the whole healthcare market
- understand the impact of the changing healthcare market
- prepare a positive healthcare environment that supports adoption of the product
- communicate the "value" of the product to the range of stakeholders that influence its introduction and adoption

This paradigm and function of MA are already under implementation in Western markets, with several companies establishing dedicated or cross-functional teams that can handle the multiple facets of MA. In emerging markets, MA is just a beginning. However, the road seems still long to go. Especially in emerging markets, MA still faces several unique challenges:

- Lack of organizational support (e.g., budgetary constraints to develop a new team dedicated to MA)
- Dearth of resources at both global and local offices
- Low level of cooperation between different functional teams (e.g., medical, marketing, etc.)
- Lack of alignment and integration of MA activities among divisions operating in different countries or markets
- Paucity of data critical to MA function (e.g., data on pricing, reimbursement, tenders, formularies, etc.)
- Identification of the right stakeholders and effective engagement with them
- Identification of areas of national importance from a policy-shaping standpoint.[1]

Many of these challenges are also common in developed or mature markets. In any case, futures for pharma and MedTech companies do not envision any change in the dynamics that brought the need for MA:

- burgeoning costs
- limited resources
- attention to performance and outcomes
- higher engagement of multiple stakeholders

For pharma and MedTech companies, this will imply that developing an MA function further is not just being a technical issue, but above all is a change in their organizational culture. Something that will challenge the way how business is conducted today. In this new environment, words such as partnerships and engagement between industry and health systems would have a new and deeper meaning.

OBJECTIVE 17.2. OVERVIEW PRICING PRACTICES OF PHARMACEUTICALS

Pharmaceutical industries and governments use different methods in pricing pharmaceutical interventions. The price of drugs keeps increasing even though governments and health insurance companies take many measures to control costs. Pharmaceutical companies tend to increase prices of marketed products.[4] However, many governments are successful in controlling or even reducing the market prices of essential drugs. For orphan drugs or even life-saving drugs, governments may subsidize partially or fully to improve the availability of the drugs in the market with affordable prices. On the other side, the cost of new drug discovery keeps rising with small success rates.

The pharmaceutical market is a highly regulated market. Pharmaceutical and healthcare markets behave differently in many aspects compared with other businesses. In many jurisdictions, competitive pricing is not allowed above or below certain levels. Ceiling price blocks prices above maximum retail price (the United States is an exception with less price regulation); and discounting below a certain percentage is not allowed to prevent cut-throat competition. Pharmacists need to understand the combination of clinical sciences with business domains. Thus, the pharmaceutical market is not "perfectly" competitive. The intention to make a profit does not overrule a clinical decision to avoid a harmful treatment even though the patient requests it.[5]

The interactions between pharmaceutical manufacturers and consumers are limited to prescription-only products, even the interactions between manufacturers and providers are monitored by governments to avoid unnecessary prescribing for profit.[6] Technically, both sellers and buyers do not have much market power. Thus, they are price takers. Increasing price loses customers, and patients are not in a position to bargain to decrease the price.[5] This is not fully true when a patented proprietary product has market exclusivity (monopoly). Still, a pharmaceutical company has restrictions by most of the governments to fix a reasonable price. In some countries with universal health coverage, the government is the largest buyer on behalf of its citizens. In this case, the government has the market power to bargain for a better price (monopsony).[7] Even with the government's control, the price keeps escalating and the expenses in innovation are high. The pharmaceutical market is strange compared with other markets in a general sense that the consumer is not the customer. Healthcare professionals are influential and, in many cases, dominate the decision on choice of the healthcare product consumed by the patient. Considering the current health insurance system, especially in developed countries, patients do not have the privilege to pay for the products and services they receive. Of course, it is not free, they did spend health taxes/insurance premiums, but in a clinical encounter, to the provider, it is a third party that pays. Above all, patients being less informed about what is going on with disease and health technology are in a poor position to contribute to the decision-making process unless they self-study or the healthcare professionals educate them. Though lots of efforts are in place to enhance patient centeredness in practice, the inherent defects explained in this paragraph limit the privileges of a customer in healthcare. It is not easy for governments to cover the health expenses of citizens, even in developed countries. Regulatory and policy reforms in the United States decreased the percentage of the uninsured.[8]

Supply and Demand Curves

Price, supply, and demand are related to each other. As quantity supplied increases, the demand decreases forcing prices to be decreased. In the reverse case, with the scarcity of the product/service supply, demand increases creating room for increasing prices (Fig. 17.2). Once the innovator drug patent expires, entry of other brands/generic versions increases the supply of the bioequivalent products in the market that decreases demand and thus the price. Having an innovator proprietary product patent is a market monopoly that creates sole demand for the product. Entry of similar products to market with smaller prices creates competition and choice easily changes to cheaper versions as the regulatory system demands similar quality, quantity, and performance across the products. A combination of demand and supply curves joins at a point of equilibrium where it defines the ideal market price and the ideal quantity to keep the demand alive. Overproduction or oversupply disturbs market equilibrium.[5] For hospital managers, bulk ordering increases supply and decreases price, but the supply should not be too much to avoid some products that are not sold before its expiry.

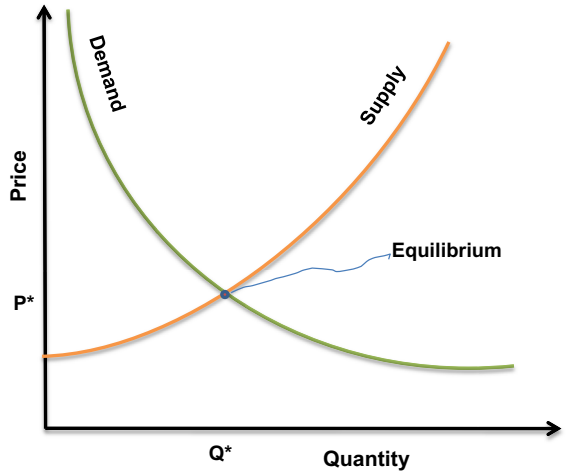


FIGURE 17.2 Supply and demand curves. *Taken from http://braungardt.trialectics.com/sciences/economics/market-mechanism-explained/.*

The price of similar pharmaceutical products (same active ingredient with same quantity) varies between countries and even within a country. Sometimes, the same manufacturer makes similar products and markets them with different brands at different prices. This variation is observed globally irrespective of high-income or low-income countries. Money spent to buy a product from the government, hospital, or a pharmacy varies depending on many factors such as purchase deal, quantity, distribution costs, etc. Deals are also in place between the providers and third-party payers or patients with certain discounts. Thus, there are variations in procurement costs for the providers and purchase price for the patients or third-party buyers.[7] Different countries might use different or a combination of methods in pharmaceutical pricing.[9]

Reference pricing is set by selecting a drug in a therapeutic class as standard price (usually of average or less than average price). The prices of other drugs with similar effectiveness and safety are compared with the standard drug. This method is commonly used by governments and other large third-party buyers. The reference price is also used in reimbursement when a patient buys the standard drug and pays the regular co-pay only. If the patient chooses a higher-priced brand, the amount above the benchmarked price needs to be paid in addition to the regular co-pay. This difference is the brand premium. A combination of reference pricing and brand premium encourages patients to choose cheaper drugs and the pharmaceutical companies to reduce the costs.[7]

In the face of rising healthcare costs across many regions worldwide, decision-makers and regulators have developed a variety of methods to control pharmaceutical and other health technology costs while attempting to maintain access to new innovative health technologies. One commonly used method of directly controlling prices is external price referencing (EPR), which is defined by the World Health Organization (WHO) as "the practice of using the prices of a pharmaceutical product in one or several countries in order to derive a benchmark or reference price for the purposes of setting or negotiating the price of the product in a given country," wherein decision-makers specify a "basket" of countries, whose prices they use to inform their national target price.[10] EPR should be distinguished from internal price referencing, where the prices of identical or therapeutically similar products are used for benchmarking purposes. EPR is frequently practiced worldwide, although countries vary substantially in how they perform EPR. Countries utilize ERP systems primarily to obtain what they believe to be fair and reasonable prices for pharmaceuticals.[11] External reference pricing is also called international price comparison.[12] Government controls the price of products at or below the average price in reference countries. Reference countries should be chosen by studying the pricing policies, production and distribution system, and even the healthcare delivery, reimbursement, taxes, etc., in the health system. These factors influence the cost and a correction should be employed if it is significantly different.[7] External reference pricing is a strategy to control the price of patented products with market exclusivity. Reference pricing practice may vary in different countries.[12]

The advantages of using EPR as a benchmark for decisions within national price control are obvious. First, it seems simple; second, it affords reassurance for most people that prices are not higher than in other countries. Additionally, EPR can be used to drive prices down and can produce cost savings, at least in the short run. Although EPR is not a perfect method, it is used commonly.[13,14]

Selecting reference countries with a similar economic status and health system objectives increases the likelihood of arriving at appropriate levels of reference pricing. Referencing countries that are of lower economic status or have unusual conditions should be avoided. Demanding the same price in lower-priced markets as in higher-income markets could cause innovative pharmaceuticals to become prohibitively expensive for developing countries. In addition, heavily referencing lower-income countries could lead pharmaceutical companies to delay launches in those countries.[14]

The wide use of EPR often has some unintended consequences internationally, which may directly or indirectly affect members of the broader stakeholder community. Worldwide decreases in drug prices may lead to decreases in research and development of new products. WHO has stated that research into innovative pharmaceuticals should be considered a global public good, which is currently driven by price alone.[10] Consequently, the value of pharmaceutical innovation to the healthcare system should always be considered and reflected in drug prices. This perspective requires that systems consider the international implications of their pricing policies. There are also concerns that EPR may cause access disparities through a price increase and launch sequencing and nonentry into certain markets.

OBJECTIVE 17.3. EXPLAIN TYPES OF HEALTHCARE COSTS AND THEIR ESTIMATION

Costing Terms

Economists should distinguish average cost from marginal cost. Average cost is the total cost divided by the number of units produced, whereas marginal cost is the change in total cost that results from the production of an additional unit. In practice, the marginal cost is usually greater than the average cost. Marginal costs can be then distinguished from incremental costs. The incremental cost is the increased cost of one healthcare intervention/program relative to an

alternative, whereas marginal cost is the change in total cost that results from the production of an additional unit. The incremental cost differs from the marginal cost in that the former relates to treatment alternatives, whereas the latter refers to more of the same treatment.[15]

Cost Categorization

All the possible sources of costs for a disease, or for increased severity of illness, should be considered to clarify the role of the new intervention that reduces harmful events of the disease. Costs are calculated to determine the resources that are used in the production of a good or service. The most basic cost of a drug or pharmaceutical product is its acquisition cost from a supplier. However, this is a small component of the overall cost. In the field of pharmacoeconomics, the medical costs can be categorized in four ways: direct medical, direct nonmedical, indirect costs, and intangible costs, to provide a comprehensive assessment.[16,17]

1. **Direct costs**: Direct medical costs refer to the costs that are directly related to the resource use associated with a service or commodity needed for providing a healthcare intervention. These are the medically related inputs used directly to provide the treatment. These include
   - **a.** Acquisition cost of the drug (drug price)
   - **b.** Transportation (shipping and insurance)
   - **c.** Supplies and equipment to administer the drug
   - **d.** Supply management (storage facilities, supply personnel)
   - **e.** Medical and allied health consultations
   - **f.** Costs of managing adverse effects of therapy
   - **g.** Hospitalization costs related to adverse events or treatment effectiveness
   - **h.** Laboratory services
   - **i.** Outpatient visits
2. **Direct nonmedical costs** comprise resources supporting the provision of healthcare services but are not medical in nature, such as
   - **a.** Travel
   - **b.** Disease-related time expenditure of affected patients and relatives providing care
   - **c.** Palliative care

$$\text{Direct costs} = \text{Direct medical costs} + \text{Direct nonmedical costs}$$

3. **Indirect costs** are costs associated with lost production capacity because of illness or death and include
   - **a.** Time lost from work for the patient (morbidity costs)
   - **b.** Time lost from work for the caregiver
   - **c.** Premature death (mortality costs)

$$\text{Indirect costs} = \text{Morbidity costs} + \text{Mortality costs}$$

The majority of pharmacoeconomic analyses focus on the direct costs of medical intervention and services provided within the healthcare system; important costs also arise outside this system. Individuals who undergo treatment are contributing their own time and thus incur an opportunity cost.[18]

In addition, individuals with the disease will tend to be less productive in the labor market and also in nonmarket uses of their time, including household tasks; productivity losses are also for those whose illnesses lead to premature death. These productivity losses represent an opportunity cost to society (and likely an income reduction to the individual) and should be considered in any assessment of the burden of disease.[18]

Indirect benefits, which are savings from avoiding indirect costs, are the increased earnings or productivity gains that occur due to medical intervention.

4. **Intangible costs** include the costs of pain, anxiety, or fatigue that occur because of a disease or its treatment. Intangible benefits, which are avoidance or alleviation of intangible costs, are benefits that result from a reduction in pain and suffering related to a medical intervention. It is difficult to measure or place a monetary value on these types of costs.

$$\text{Total costs} = \text{Direct costs} + \text{Indirect costs} + \text{Intangible costs}$$

Intangible costs are usually incorporated in the benefits side and incorporated as utilities assigned to health states that reflect the quality of life (QoL). QoL measures a person's experience of life, including work, recreation, social activities, etc. The impact of a person's health status on their QoL, i.e., their health-related QoL is usually measured in pharmacoeconomic analysis.[19]

Standardized questionnaires measure either the health-related QoL in general or QoL of a specific disease. Short Form 36 (SF 36) and EuroQol tools (EQ-5D or EQ-5D-L) are widely used generic health-related QoL instruments. SF 36 has eight domains: physical functioning, social functioning, role functioning, psychological distress, general health perceptions, bodily pain, vitality, and psychological well-being.[20]

Perspective

The perspective of a study must be determined to specify what costs are important to measure. Perspective is a term that describes which costs are needed based on the purpose of the study. Depending on the group/member making the reimbursement/payment, the following perspectives can be considered: (1) societal, (2) healthcare system, (3) third-party payer, (4) business, (5) government, and (6) participants and families. Common perspectives used in pharmacoeconomics analysis are the perspective of the payer (e.g., insurance company) and societal. The societal perspective is the broadest; it is difficult and time-consuming to estimate all of its cost components. E.g., societal costs include costs to the payer, costs to the provider/institution, other sector costs, costs to the patient, and indirect costs because of the loss of productivity.[21]

When determining costs, it is important to be evaluated in each perspective, irrespective of interventions, therapeutic areas, and patient groups investigated. In pharmacoeconomics, costs are considered from various perspectives (Table 17.1).

Estimating Costs

As a principle, costs should be determined as precisely as possible, including the methods, sources, and results in individual steps of cost estimations. The estimation of costs follows a process:

1. ***Identification of resources*:** Within the framework of the resources identified for the provision of healthcare services used for treating the disease, the information can be obtained from different sources. The sources are primary data (individual items of a particular service, e.g., drugs, laboratory, radiology, etc.), secondary data (e.g., health insurance data, government surveys), guidelines (e.g., WHO datasets), and expert opinions.
2. ***Quantification of resources*:** The frequency of use, the proportion of the relevant patient population using each service, and the duration of the service must be determined. The measuring or quantification of resources could be amounts of labor input or outputs (e.g., bed days, time in theater, prescriptions, etc.) and could include patients' or carers' time. Costs of service that are infrequently used and have only a slight impact on the results should be described, but not needed for the calculation. The costing method can use a micro- or the macrocosting approach. The degree of precision of quantification is determined by the perspective and reimbursement system, and the corresponding degree of aggregation of the services.

TABLE 17.1 Perspective and Relevant Costs to Be Considered

Perspective	Medical Costs	Morbidity Costs	Mortality Costs	Transportation/ Nonmedical Costs	Transfer Payments
Societal	All costs	All costs	All costs	All costs	
Healthcare system	All costs				
Third-party payer	Covered costs		Covered costs		
Businesses	Covered costs (self-insured)	Lost productivity (presenteeism/absenteeism)	Lost productivity		
Government	Covered (Medicare, Medicaid)			Criminal justice costs	Attributable to illness
Participants and families	Out-of-pocket costs	Lost wages/household production	Lost wages/household production	Out-of-pocket costs	Amount received

3. ***Evaluation of the resources*:** If there are prices of healthcare services and if these can be assumed to reflect costs, these can be multiplied by the relevant units of service use to yield total costs, such as x days multiplied by USD per day. For a detailed evaluation of resources, microcosting (bottom-up method) or gross costing (top-down method) is used. For cost identification, in microcosting, all cost components are defined at the most detailed level, whereas in gross costing, cost components are defined at a highly aggregated level. For cost valuation, in the bottom-up approach, cost components are valued by identifying resources used directly employed for a patient, whereas in the top-down approach, cost components are valued by separating out the relevant costs from comprehensive sources.[22] Although analysts favor microcosting, it tends to be costly and runs the risk of being specific to particular contexts. Top-down costing allocates a total budget for specific services such as hospital stays or doctors' visits according to rules. Bottom-up microcosting is considered to overestimate assessed costs, top-down gross costing tends to underestimate them. The choice between microcosting and gross costing depends on the needs of the analysis. Many studies use a mixture of the two, using microcosting for the direct costs of the intervention and gross costing for other costs.[22] Combining these dimensions gives four theoretic approaches: top-down gross costing, top-down microcosting, bottom-up gross costing, and bottom-up microcosting.[35]

Activity-based costing (ABC) is a costing methodology that identifies activities necessary to produce a defined service or product and assigns the cost of each activity resource to processes according to the actual consumption associated with the provision of activity. Models can be implemented relatively quickly because only estimates of two parameters are required: (1) the unit cost of supplying capacity (e.g., cost per minute of a surgeon's time) and (2) the time required to perform a transaction or an activity. The breakthrough of time-driven activity-based costing (TDABC) lies in using time equations to estimate the time spent on each activity.[23] TDABC helps to perform costing in health centers efficiently. It is intended to capture the cost of care delivery for conditions accurately and to control cost in an effort to create value in healthcare.[24] TDABC is more accurate and simpler than traditional ABC because it assigns resource costs to patients based on the amount of time clinical and staff resources are used in patient encounters.[25]

Methods to Measure Indirect Costs

Methods of measuring indirect costs include human capital, friction cost, and willingness to pay (WTP). The human capital method involves factors related to the production of valuable goods or services. Efforts of a talented workforce result in required productivity. It is the investment of a trained workforce that results in better productivity. Productivity losses of an individual due to morbidity or mortality are valued toward the present value of his or her future earnings. Assumptions about future earnings could be wrong. Considering that a worker cannot be replaced even when the employment rates are high is also debatable. It results in an overestimation of indirect costs due to productivity losses. The friction cost method, on the contrary, considers the productivity loss to be avoided when another person is employed and trained to the same talent levels. It assumes that morbidity or mortality will not affect total productivity. The time taken from the discontinuation of the previous employee due to ill health or death until a new employee is trained to comparable productivity is called the friction period. The method is controversial for considering morbidity and mortality to reduce unemployment. The human capital and WTP methods are commonly used.[26]

The WTP method can value both the indirect and intangible aspects of a disease or condition. The WTP method determines how much people are willing to pay to reduce the chance of an adverse health outcome. To elicit WTP values, respondents are presented with a hypothetical scenario describing the benefits of a particular healthcare intervention (e.g., program, pharmaceutical, medical device). Respondents are then asked to value the healthcare intervention in a dollar amount.[11]

A description of the healthcare service or intervention (e.g., medication therapy management program, new drug therapy) should be included in the hypothetical scenario. The scenario intends to provide the respondent with an accurate description of the good or service that he or she is being asked to value. In addition, the scenario should detail the amount of time the person should expect to spend, as well as the benefit (e.g., percent improvement in the condition) of the intervention. After the program or intervention has been adequately described, respondents are then asked to "bid" or place a value on the program or intervention. Bids can be obtained through a variety of formats, such as open-ended questions, closed-ended questions, a bidding game, or a payment card.[27]

1. Open-ended questions, simply ask respondents how much they would be WTP for the program or intervention.
2. Closed-ended questions, respondents are asked whether or not they will pay a specified dollar amount for the program or intervention.
3. Bidding game resembles an auction in that several bids are offered to reach a person's maximum WTP.
4. Payment card provides the respondent with a list of possible WTP amounts (i.e., payment card) to choose from.[27]

Time Adjustment for Cost

Standardization of Cost

If retrospective data are used to assess resources used over the years, more than 1 year before the study these costs should be adjusted before a direct (fair) comparison between medical interventions. Standardization is carried out by calculating the number of units (doses) used per case and multiplying this number by the current unit cost for each medication.[28]

Another method for standardization is to multiply all of the costs from the year the data were collected by the medical inflation rate for that year. Medical Consumer Price Index inflation rates can be found at the Bureau of Labor Statistics' website (www.bls.gov) and have been between 3% and 4% each year since 2005.[29]

Discounting

The time horizon is used to determine a period of time during which the outcomes of analysis will be considered. The time horizon could be expressed as a fixed number of years (or months or weeks) or relative to study variables (e.g., patients' lifetimes). In other words, the time horizon is a point in the future up to which all costs and effects must be accounted for and beyond which everything can be ignored.[30]

If the time horizon of a pharmacoeconomic analysis is several weeks or months, no discounting is required. However, if the time horizon is several years, then costs that occurred at different times must be brought to the same reference time point.[31]

Discounting is needed when costs are determined based on dollars spent or saved in future years. The discount rate is used to estimate the time value of money or benefits. The generally accepted discount rate for any healthcare interventions is between 3% and 5%, but it is recommended that the discount rate should be varied using high and low estimates of various discount rates.[32] To study the effects of varying discount rates on results of the study, a sensitivity analysis could be performed.[33]

The discount factor is equal to $1/(1 + r)^t$, where r is the discount rate and t is the number of years in the future that the cost or savings occur.[34]

CONCLUSION

Pharmaceutical pricing is different from other industries for many reasons. Different types of governmental control regulate the market. Healthcare costs are both direct and indirect. Identifying and valuing costs accurately have challenges, especially with indirect costs. Best practices in estimating costs are essential to build on evidence in value-based decisions.